Major Project (BTCS706N) - Weekly Progress Report

Project Title: Quantum Secure Email Client Application (QuMail)

Team ID: G14

Reporting Period: Week 4

Phase: Phase 2 – Object-Oriented Analysis

#Executive Summary

Current Sprint: Sprint 4 of 6

Sprint Goal: Continue Phase 2 – Object-Oriented Analysis with a focus on Dynamic Analysis. The objective is to model how the QuMail system responds to user actions and system events using an Activity Diagram.

Status: On Track

During Week 4, the dynamic workflow of the Quantum Secure Email Client was modelled. The Activity Diagram represents user authentication, email composition, encryption-key requests, email encryption, sending through the email server, receiving, decryption, and reading the message. The Week 3 Object Diagram and Initial Class Diagram serve as references for this workflow.

#UML & OOAD Progress

Diagram prepared: Activity Diagram — models the dynamic flow of secure email communication, including activity sequence and the authentication decision point.

Earlier diagrams referenced:

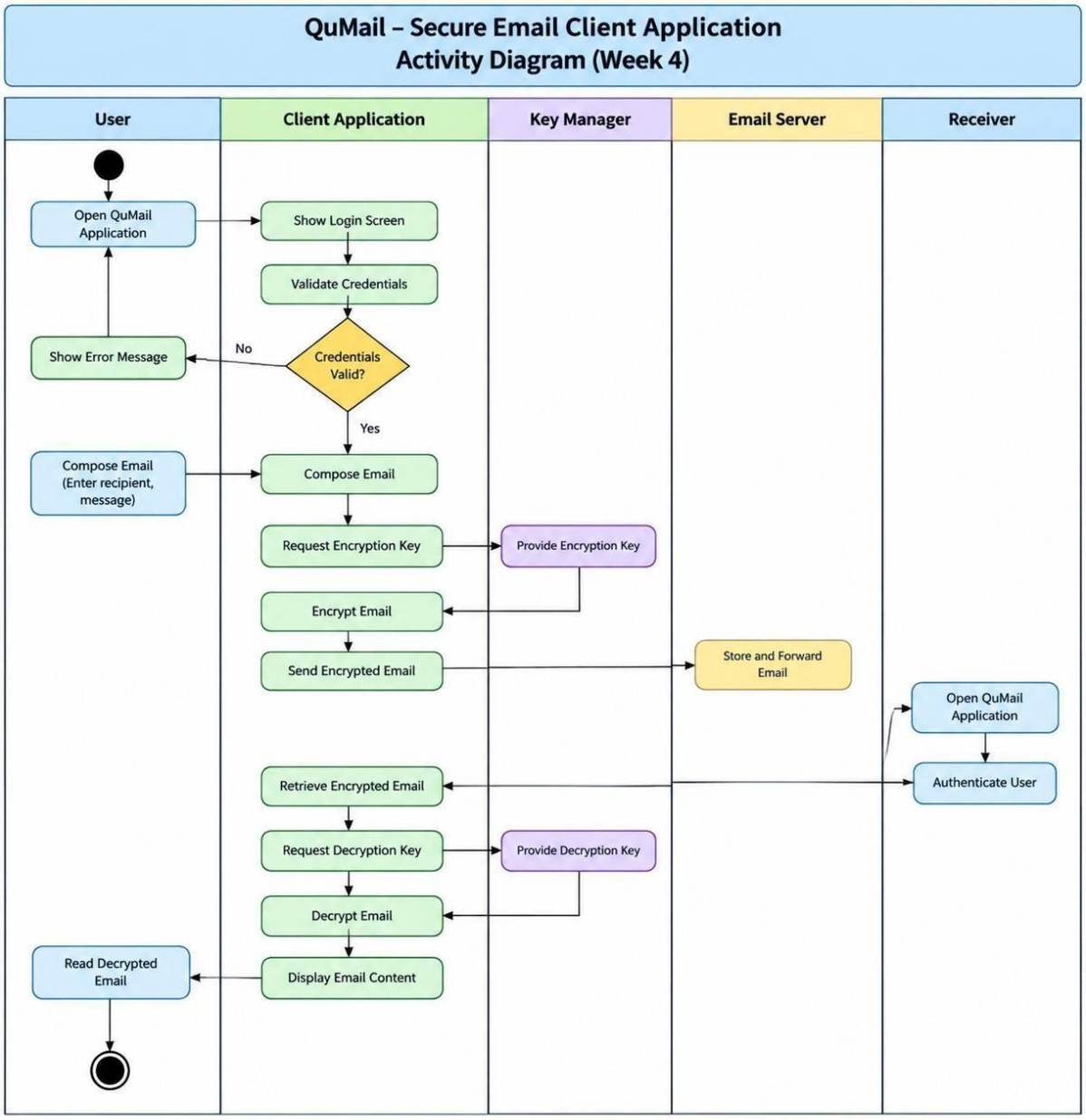
- Object Diagram — represents specific instances and a data snapshot of the system.
- Initial Class Diagram (Domain Model) — identifies domain classes, attributes, and relationships without methods.

OO Concept Applied: Dynamic analysis represents how system activities proceed and how the workflow branches depending on whether the user's credentials are valid. The model connects the client application, Key Manager, Email Server, sender, and receiver.

#Activity Diagram Overview

The next page contains the Activity Diagram for the secure-email workflow, from opening QuMail through authentication, encryption, transmission, retrieval, decryption, and reading.

UML Deliverable — Activity Diagram



#Technical Execution & Testing Summary

Technical Execution (DevOps & Git)

Jira / Project Tasks:

- Dynamic-analysis work for Phase 2 was addressed through the secure-email activity workflow.
- Activities for authentication, encryption, sending, receiving, and decryption were identified in the diagram.
- The Activity Diagram was prepared for inclusion in the Week 4 project documentation.
- Jira task: SCRUM-18 (Activity Diagram).

Git Activity:

- The Activity Diagram and report were saved in the project folder.
- The documentation changes were committed and pushed to the repository.

CI/CD Status: CI/CD implementation is not the primary focus of this report. Week 4 focuses on Dynamic Analysis and UML Activity Diagram modelling.

Testing Summary

Total Unit Tests: 0

Pass Rate: N/A

Validation: Review the activity sequence, decision branches, and links between sender, client, Key Manager, Email Server, and receiver.

Key Bug Identified: None. This week focused on UML modelling; no implementation bugs were observed.

Goals for Next Week

- Review and refine the Activity Diagram based on project-guide feedback.
- Check consistency between Week 3 static models and the Week 4 dynamic workflow.
- Continue the next project activities according to the approved project plan.
- Maintain UML documentation and project changes through Git.

Mentor Signature:_____ Project Guide Signature: _____